Quebrada Azul Copper Expansion
Client: Bramhall Copper Corp.
Contractor: Ostrander Engineering Services Ltd.
Project number: 7-0112

GENERAL SITE CONDITIONS

Document number: QAZ-EP-0000-210-DBD-00002-B
Revision: B
Date: 2019-10-02
Discipline: 210
Status: Issued for review
Originator: Mireille Athey, Site Engineer
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This document constitutes the environmental and site-data engineering basis for the Quebrada Azul Copper Expansion. It shall be interpreted alongside the specification and data sheets enumerated within Section 1.4, and its stipulated values govern wherever those referenced publications remain silent.

Cover notes

The location data herein define the atmospheric, climatic and geotechnical subsurface conditions which every engineering discipline shall apply when dimensioning, rating and verifying installations.

Design magnitudes stated herein are engineering design parameters exclusively. They are not performance guarantees regarding any equipment or installation.

Document register and revision history

The register within Section 1 documents every revision of this document chronologically by publication, including its date, purpose of publication, originator, checker, approver and document controller. The prevailing revision appears finally and supersedes all preceding issues. Dash-marked rows are reserved for prospective revisions and contain no issued information. A revision is authoritative exclusively when recorded within this register and promulgated through the Document Controller.

Rev	Date	Description	Originator	Checker	Approver	Doc. controller	Status	Notes	Sheets
A	2019-09-06	For information	Mireille Athey	None	Simon Veil	Paulo Maren	Superseded	Superseded; for information only	9
B	2019-10-02	Issued for review	Mireille Athey	None	Simon Veil	Paulo Maren	Current	Current; issued for review	9

Document conventions

Dates herein are expressed according to YYYY-MM-DD formatting. Units are consistently SI, with derived units presented in complete symbolic notation. Altitudes and elevations are expressed in metres above sea level, designated masl following the magnitude.

Magnitudes are stated without tolerance unless the corresponding tabulation specifies one. Percentages are typeset without any whitespace preceding the percentage symbol. Negative temperatures employ a hyphen, and a solitary space separates every numerical magnitude from its corresponding unit.

Abbreviations

Abbreviation	Meaning and use in this document
DBD	Design basis document governing engineering criteria
GMD	Gearless mill drive for grinding mill propulsion
LAHJ	Local Authority Having Jurisdiction over approvals
LV	Low voltage distribution for auxiliary equipment
MCC	Motor control center housing starter assemblies
MV	Medium voltage distribution for primary switchgear
NAEC	Norte Alto Electrical Code governing installations
OLTC	On-load tap changer regulating transformer voltage
ONAF	Oil natural, air forced cooling for transformers
ONAN	Oil natural, air natural cooling for transformers
PET	Potential evapotranspiration for hydrological balance
PGA	Peak ground acceleration for seismic structural design
RH	Relative humidity characterizing atmospheric moisture
masl	Metres above sea level denoting site elevation
rev	Revision identifying document issue status

Basis of the site data

The location data are compiled from the site meteorological archive and from regional climatological statistics encompassing the province. They are re-examined at every revision and amended wherever the archive has been extended or rectified.

Extreme magnitudes represent the maximum or minimum observations within the site archive. Design magnitudes may incorporate an allowance beyond an extreme, as each tabulation specifies. Wind engineering employs the gust velocity specified within Section 2.2, not the average wind velocity.

1. Introduction

1.1 Purpose and scope

This document establishes the environmental and location conditions governing the Quebrada Azul Copper Expansion, project 7-0112. It applies to every discipline designing, rating or verifying installations at the location.

This document furnishes location information exclusively. Equipment ratings are not established herein; they are defined within the specifications and datasheets. Each designer shall determine the applicable rating from the site information presented.

1.2 Location of the Works

The site resides within Norte Alto Province, neighbouring the township of Calderón Verde, at an altitude of 3,850 masl and 620 km from the regional capital. The following tabulation enumerates the geographical parameters mandated for engineering design.

Location item	Value
Province	Norte Alto Province
Nearest town	Calderón Verde
Distance to regional capital	620 km
Altitude	3,850 masl

1.3 Access to the Site

Site accessibility is by roadway from the regional capital. Loading restrictions apply along the concluding route segment, and equipment mobilisation for all heavyweight and over-width consignments shall be scheduled accordingly.

Abnormal consignments necessitate a route reconnaissance encompassing bridges, gradients, turning radii and overhead clearances before transportation is reserved.

1.4 Related Documents

Interpret alongside GMD Specification QAZ-EP-0000-265-SPC-00006-Q, GMD Data Sheet QAZ-EP-0000-265-DST-00006-Q, Grinding Mill Data Sheet QAZ-EP-3100-255-DST-55017-Q together with Single Line Diagram QAZ-5200-265-E-001_B.

Any discrepancy between this document and a project document shall be communicated to the Document Controller before engineering commences.

1.5 Reference Codes and Authorities

Electrical engineering shall comply with the Norte Alto Electrical Code NAEC-2019. The Local Authority Having Jurisdiction regarding the Norte Alto Province administers the regulations and grants electrical installation authorisations. The Contractor shall coordinate every LAHJ-mandated examination preceding energisation.

2. Environmental and Site Data

2.1 Air Temperature

Atmospheric temperatures are tabulated below. The design ambient is 3 °C above the extreme maximum and is employed for ratings.

Parameter	Value	Basis
Minimum temperature	-12 °C	Extreme minimum in site record
Maximum temperature	27 °C	Extreme maximum in site record
Design ambient temperature	30 °C	3 °C above extreme maximum; for rating

2.2 Wind Speed

The design wind velocity is the tabulated gust magnitude. It pertains to structures, enclosures and all unsheltered equipment.

Parameter	Value	Basis
Design wind speed	36 m/s	3-second gust at 10 m above ground
Gust averaging period	3-second	Gust basis, not mean wind
Reference height	10 m	Measured above ground level

2.3 Solar Irradiance

Solar irradiance values below are utilised for thermal loading on enclosures and cooling installations, and for thermal verifications of exposed equipment.

Parameter	Value	Basis
Peak global irradiance	1,050 W/m²	Highest value in site record
Daily average irradiation	6.2 kWh/m²/day	Average over the year
Data basis	Site meteorological record	Site meteorological record
Use	Cooling and enclosure heat load	Enclosure and cooling heat load
Design check	Thermal checks	Thermal check of exposed equipment

2.4 Humidity Conditions

RH magnitudes are utilised for enclosure and cooling engineering. The maximum is employed to evaluate condensation and insulation.

Parameter	Value	Basis
Mean relative humidity	58%	Annual mean
Maximum relative humidity	95%	Highest value; condensation, insulation

2.5 Evapotranspiration Demand

PET is utilised within the site hydrological balance together with the average annual precipitation presented within Section 2.7.

Parameter	Value	Basis
Annual potential evapotranspiration	1,320 mm/yr	Annual total
Basis of estimate	Site meteorological data	Derived from site climate record
Use	Site water balance	Water balance, supply and storage checks

2.6 Seismic Parameters

The location occupies seismic zone 4, with a design peak ground acceleration of 0.42 g on site class D. These parameters govern seismic restraint of mill drivetrains, transformers and enclosures, together with foundation anchorage detailing.

2.7 Annual Rainfall

Average annual precipitation at the location is 410 mm/yr. This magnitude governs site drainage engineering, encompassing ditches, sumps and retention-pond dimensioning. Outfalls and discharge locations shall be verified against the PET magnitude of 1,320 mm/yr within Section 2.5 throughout hydrological water-balance preparation.

Because PET exceeds precipitation, the hydrological balance is closed on a net deficit basis, and drainage is dimensioned for stormwater runoff, not for surplus inflows.

2.8 Depth of Frost Penetration

The design frost penetration is 450 mm beneath finished ground level. Foundations, pipework, cabling and other subterranean services shall be installed beneath this threshold. Where frost heave threatens structural uplift or fracturing, the designer shall specify sufficient embedment depth or thermal insulation.

Where subsoil is frost susceptible, the designer shall verify it and substitute or safeguard it as necessary.

3. Attached Information

The documentation appended to this document constitutes part of the site information and shall be utilised for design verification together with Section 2.

Site location drawing illustrating Norte Alto Province, the township of Calderón Verde and the highway access route from the regional capital to the location.

Climatological summary presenting the monthly magnitudes that substantiate the information within Section 2 for each parameter, together with the duration of the site archive utilised.

Extract of the revision register for this document as it existed at this issue, illustrating Rev A, Rev B and the personnel responsible.